Exercise

100 $(1-\alpha)\%$ C.I for ρ:

$$P\left(-Z_{\frac{\alpha}{2}} < \frac{\frac{1}{2}\log\left(\frac{1+r}{1-r}\right) - \frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right)}{\sqrt{\frac{1}{n-3}}} < Z_{\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(-Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}} < \tanh^{-1}(r) - \tanh^{-1}(\rho) < Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right) = 1-\alpha$$

$$P\left(-\tanh^{-1}(r) - Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}} < -\tanh^{-1}(\rho) < -\tanh^{-1}(r) + Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right) = 1-\alpha$$

$$P\left(\tanh^{-1}(r) - Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}} < \tanh^{-1}(\rho) < \tanh^{-1}(r) + Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right) = 1-\alpha$$

$$P\left(\tanh\left(\tanh^{-1}(r) - Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right) < \rho < \tanh\left(\tanh^{-1}(r) + Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right)\right) = 1-\alpha$$

$\therefore$ 100$(1-\alpha)\%$ C.I for ρ:

$$\left(\tanh\left(\tanh^{-1}(r) - Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right), \tanh\left(\tanh^{-1}(r) + Z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right)\right)$$ #